

Architectural Katas 2026: AI-Assisted Software Architecture



What Is A Kata?



CodeKata

*Because experience
is the **only** teacher*



[PragDave](#) [Kata](#) [Archives](#)

CodeKata

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How do you get to be a great musician? It helps to know the theory, and to understand the mechanics of your instrument. It helps to have talent. But ultimately, greatness comes from practicing; applying the theory over and over again, using feedback to get better every time.

How do you get to be an All-Star sports person? Obviously fitness and talent help. But the great athletes spend hours and hours every day, practicing.

But in the software industry we take developers trained in the theory and throw them straight in to the deep-end, working on a project. It's like taking a group of fit kids and telling them that they have four quarters to



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Architectural Katas

"How do we get great designers? Great designers design, of course." --Fred Brooks

"So how are we supposed to get great architects, if they only get the chance to architect fewer than a half-dozen times in their career?" --Ted Neward

[Do one! »](#)

About

The Architectural Katas started as a presentation workshop by Ted Neward. They've taken on a life of their own.

[Learn more »](#)

Invite

Want an experienced Architectural Kata moderator to run the workshop at your place of business?

[Contact »](#)

Rules

Doing an Architectural Kata requires you to obey a few rules in order to get the maximum out of the activity.

[Read rules »](#)

Lead

Want to run the Architectural Katas yourself? There's only a few things you need to know before you do.

[Learn how »](#)

Contribute

New Kata problems/proposals are always welcome.

[Send ideas »](#)

What do we want you to do?

You'll be presented with a realistic example of a company who want you to define an architecture for them

We want you to come back with your proposal for how the system should be built

We want you to pay special attention to how the use of AI tooling could help enhance the solution, both for the company, and also for the companies customers.

The Problem



Background

After a bizarre gardening accident, the 204th in line to the Von Digitalis estates has suddenly found themselves the newly appointed 72nd Countess Von Digitalis

The estates are large and sprawling, and are in need of some help to make them profitable

The family's previous business of making unfortunately highly explosive garden gnomes has proved to no longer be viable

The 72nd Countess Von Digitalis is looking to bring some digital solutions to help her monetize parts of the estate



There is an extensive, historically important collection of 18th century amusement park rides on the estates too.

The rides have recently passed safety inspections after all the asbestos, broken glass and garden gnomes were removed

Photo by [jim gade](#) on [Unsplash](#)

The previously private exotic and poisonous animal collection is also going to be opened to the public

There is a mix of aquatic and land-based animals

Photo by [David Clode](#) on [Unsplash](#)



Context

The Von Digitalis Estates get on average get 5000 visitors a day, with expectation (and hope!) that this will grow to at least 15000 within three years, or else the family may be forced to sell their carnivorous plant collection

There are 40 rides in the amusement park area, and over 200 animals in exotic animal collection split across 55 different displays and enclosures

We want you to come up with a new
comprehensive modern architecture for the
Von Digitalis estates

What We Need!

We need a system that provides the ability for people to buy tickets, including family passes to access the estates

We want to understand how popular different parts of the park are, so we can understand where we need to improve things

The exotic animal collection also requires careful monitoring - we want ways of tracking animal health, how much/well they are eating, and in the case of the jumping pirana collection, we also need to check population levels

We also need to work out how to grow the number of visitors, but also make the estates more profitable, or it's back to the garden gnome business

Technical Context/Constraints

Wifi coverage on the park is patchy

You can make use of cloud services, but you need to have some ways of getting information from the estate to the cloud

Assume there is a budget for MQTT-capable hardware devices which can be installed throughout the park

Biggest business challenges

We have no real idea of what parts of the estates are most popular, so it's difficult to know where to invest & deploy staff

Looking after the animals is costly, even more so if they get sick! So we want healthy and happy animals

We want to get more returning visitors too, but aren't sure how

We want you to place a focus on how AI
could be used to solve the problems of the
72nd Countess Von Digitalis

Things we expect to see....

Deliverables

- Overview: a *short* narrative describing how the team used AI to solve the problems of the Von Digitalis Estates
- Diagrams: comprehensive and targeted views for each use of AI
- ADRs for AI-related implementations, including trade-off analysis
- (optional) Pertinent implementation details
- (for semi-final teams) Five-minute video describing the team's approach

Judges Criteria

- Innovative use of AI in solution(s)
- Suitability of the solution given the constraints
- Provide appropriate levels of detail
- Dealing with uncertainty in the world of AI technology
- Do the architectural characteristics of the additions match the existing architecture?
- Validation and verification of AI results

Our judges can't speak to
you individually...

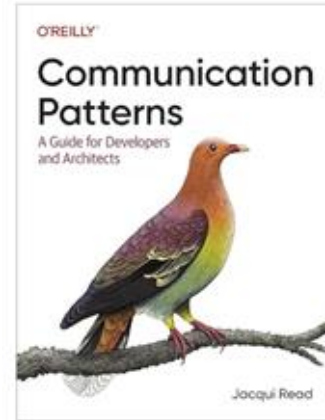
...so you need to
communicate with them via
your deliverables

Communication is key

diagrams - types, level of detail, completeness



“Successful communication is the art and science of sharing or exchanging ideas and information,... resulting in shared understanding.”
Jacqueline Read



www.oreilly.com/library/view/communication-patterns/9781098140533/

Document your
decisions!



architecture decision records

short text file; 1-2 pages long, one file per decision
markdown, textile, asciidoc, plaintext, etc.

Date: ____/____/____

■ # Title

■ ## Status

■ ...

■ ## Context

■ ...

■ ## Decision

■ ...

■ ## Consequences

■ ...

short noun phrase

proposed, accepted, superseded

description of the problem and alternative solutions available (documentation)

decision and justification (the "why")

trade-offs and impact of decision

Judges Criteria

diagrams - types, level of detail, completeness



“The goal of a diagram is to convey a clear and shared understanding of the architecture”

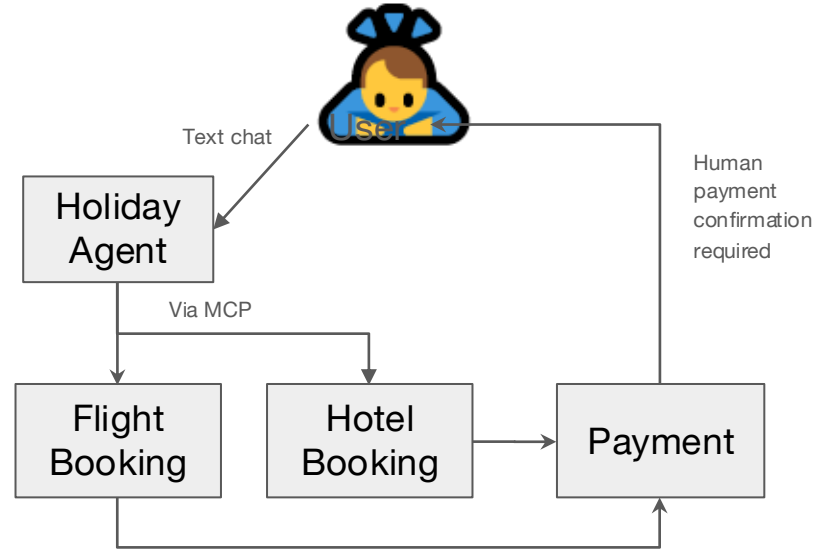
- Neal Ford

Pictures!

We don't need complex diagrams,
we need easy to understand
diagrams

You can go a long way with boxes,
arrows and words

If you use different shapes to mean
different things, please provide a
key!



Dealing with uncertainty in AI

AI is changing FAST - how are you going to deal with the fact that the best models or providers today might not be the best tomorrow?

How would you handle your model provider changing prices on you?

What might happen if the provider you used suddenly shut down?

Does it work?

We want to understand how you will confirm that your use of AI is actually working

Verifying functionality of deterministic solutions is easy - but GenAI functionality tends to be non-deterministic!

How will you know if your AI-driven functionality starts misbehaving once in production?

Meet the Judges

Gayathri Mohan

Gayathri Mohan is an independent technology consultant who helps organizations modernize quality engineering through AI-driven test automation and stronger CI/CD practices. She has over 17 years of experience, including 15 at Thoughtworks, where she helped shape global AI strategy, led global capability-building for quality engineering, and built out the Chennai office as the AI/ML hub. She's also the author of the book [*Full Stack Testing*](#), coauthor of *Perspectives on Agile Software Testing*, has written for Martin Fowler, and speaks internationally at conferences, including her recent keynote at JaSST'26 Tokyo.



Ben O'Mahony

Ben O'Mahony is a results-driven AI and engineering leader with a track record of building high-performing teams and shipping business-critical AI, ML, and data platforms at scale. He has deep expertise across the full engineering and data lifecycle, from research to production deployment and is adept at defining technical strategy, driving execution, and partnering cross-functionally to deliver measurable impact. He's currently writing [*Building AI Agent Platforms*](#) for O'Reilly, due to be published next spring, and available to read now in early release on the O'Reilly platform.



Ian Cooper

Ian Cooper is a polyglot coding architect with over 20 years of experience delivering solutions in government, healthcare, finance, and ecommerce. During that time he has worked for the DTI, Reuters, Sungard, Misys, Beazley, Huddle, and Just Eat Takeaway, delivering everything from bespoke enterprise solutions and “shrink-wrapped” products for thousands of customers to SaaS applications for hundreds of thousands of customers. He’s an experienced systems architect with a strong knowledge of OO, TDD/BDD,DDD, EDA, CQRS/ES, REST, messaging, design patterns, architectural styles, ATAM, and Agile engineering practices. He’s a frequent contributor to OSS and is the owner of <https://github.com/BrighterCommand>.





If you want some
inspiration?

The Katalog

<https://github.com/thekatalog>



How To Submit Your Problem

Create a GitHub repo for your problem and submit it on the next slide

Please include all documentation, visuals in the repo!

Make sure to have a simple readme for the repo so our judges can navigate what you have shared with us

Kata Schedule

- All teams must submit this Google Form (<https://forms.gle/trWJ39mA556vjaYPA>) by 12pm ET, Wednesday, September 2
- Solutions are due in your GitHub repo by 11:59pm ET Wednesday, September 16
- Semifinalists will be announced at the second event on Monday, October 5
- Semifinalist video presentations due by 11:59pm ET Monday, October 12
- Winners are announced at the final event on Wednesday, October 21